

An Honest Residual Image for Strong-Lens Vetting: From a Gaussian High-Pass to a Signed, Noise-Normalized Elliptical-Model Residual in LENSJUDGE

Greg Benson*

Xiaosheng Huang†

June 2026

Why the old residual could both fabricate and hide arc flux, what replaced it, how it was validated, and what a controlled A/B shows it does (and does not) do to grading

Abstract

The LENSJUDGE vetting cascade shows a Claude grader a small set of rendered views of each candidate’s DESI *grz* cutout; the “residual” view — the smooth lens-galaxy light removed so that lensed arcs stand out — is the most error-prone to compute and one of the most influential, since it is returned to the model on *every* grading call. The original residual was a per-band Gaussian high-pass, re-floored to non-negative and shown as an arcsinh RGB. We show that this transformation is *not honest to the raw data* in four concrete ways: it destroys the sign of the residual (hiding over-subtraction), it imprints a four-lobe “butterfly” quadrupole on elliptical galaxies that mimics arcs, it carries no noise normalization, and its kernel scale is comparable to the seeing so it subtracts real arc flux and rings every compact source. We replace it with a *signed, noise-normalized, elliptical-model residual*: per band, a sigma-clipped median over elliptical annuli (matched to the galaxy’s ellipticity and position angle) is subtracted from the seeing-convolved data without deconvolution, the residual is divided by a robust per-band noise σ , and the dimensionless $\chi = (\text{data} - \text{model})/\sigma$ is displayed on a fixed $\pm 5\sigma$ red–blue diverging scale as a $g|r|z$ montage. Nothing is re-floored or clipped, so over-subtraction stays visible; the median rejects a minority-fill arc so it survives in the residual; cross-band coherence becomes the discriminant between a real arc and a per-pixel artifact. We document the design choices (median model as the default, with a photutils elliptical-isophote engine as an opt-in — it converges on under 15% of these faint cutouts; and the median over a more flexible harmonic fit that would *absorb* real rings), and validate the new residual on synthetic galaxies: outside the inner $\sim 2''$ the residual is consistent with $\mathcal{N}(0, 1)$, an injected arc is recovered at 0.63 of its in-mask flux versus 0.08 for the old high-pass, the smooth-galaxy residual is $7\times$ smaller, and deliberate over-subtraction is plainly visible. The honest cost is a central PSF-circularization residual confined to the inner $\sim 2''$, shown faithfully as a blue core rather than hidden. A controlled grading A/B on 525 candidates then measured the downstream effect: the new residual is *discrimination-neutral* (lens-vs-random $\Delta\text{AUC} = +0.021$, 95% CI $[-0.04, +0.09]$); it exposed and let us fix a real regression in the grader’s residual *convention* (an early “blue = artifact” wording made the grader reject real, blue arcs); and because the new image makes the grader score $\sim 2\times$ hotter, it must be deployed through an operating-point calibration, not its raw grade. The gain is integrity and defensibility, not grading accuracy.

*benson@cs.usfca.edu; Department of Computer Science, University of San Francisco, CA 94117, USA

†Department of Physics & Astronomy, University of San Francisco; and Physics Division, Lawrence Berkeley National Laboratory

1 The residual view, and why it must be honest

LENSJUDGE grades a strong-lens candidate from a handful of rendered views of its 101-px *grz* cutout ($0.262''/\text{px}$). The Claude SDK has no image-in-prompt channel, so these PNGs — returned by the `fetch_cutout` tool — are the *only* way the model sees pixels. Three views are shown by default: the full Lupton-RGB cutout, a $2.5\times$ center zoom, and the *residual*, in which the smooth deflector galaxy is removed so that a faint lensed arc or Einstein ring can stand out against the much brighter foreground. The residual is therefore not decoration: it is evidence the automated grader reasons over on every call, and it also appears as a figure panel in the published candidate galleries.

A residual image is honest to the raw data when it *introduces no feature the data lacks and hides none that exist*. Concretely this means: it preserves the sign of the residual (excess flux and over-subtraction are equally informative), it carries a meaning in units of noise (so “how significant” is legible), it subtracts a defensible model of the smooth galaxy rather than an arbitrary spatial filter, and it does not manufacture arc-like structure where the sky has none. The original LENSJUDGE residual violated all four of these. We diagnose the failures (§2), describe the replacement (§3), compare them on real cutouts (§4), justify the design (§5), and validate it (§6).

2 The old transformation and its four failures

The original residual (`render.residual_legacy`, retained for reproducibility) was, per band $b \in \{g, r, z\}$, a difference of the data and an isotropic Gaussian-smoothed copy, then re-floored to non-negative and rendered as a reduced-stretch Lupton arcsinh RGB (Lupton et al., 2004):

$$\text{res}_b = \text{data}_b - \mathcal{G}_{\sigma=3}[\text{data}_b], \quad \text{res} \leftarrow \text{res} - \min(\text{res}).$$

This has four distinct problems, each of which is either a hidden truth or a fabricated one.

(i) Sign destruction. The step $\text{res} \leftarrow \text{res} - \min(\text{res})$ maps the most negative pixel to zero, so every pixel where the model exceeded the data — i.e. every *over-subtraction* — becomes a positive value indistinguishable from genuine excess. A too-aggressive model can then delete real arc flux with no visible trace. Over-subtraction is not a corner case: it is exactly what happens at the bright galaxy core, and the re-floor renders it invisible (Fig. 5, left).

(ii) The “butterfly” quadrupole. A deflector is a relaxed elliptical galaxy, but an isotropic Gaussian smoothing is circular. Subtracting a circular model from an elliptical galaxy leaves a four-lobe quadrupole: a positive excess along the major axis (which *mimics a tangential arc*) and a negative trough along the minor axis (which *erases* real flux there). Figure 1 shows this directly on a synthetic elliptical galaxy. This is the most insidious failure: the artifact looks like the signal.

(iii) No noise normalization. The output is in arbitrary counts passed through an arcsinh stretch. Two consequences follow. First, the viewer (human or model) cannot tell a 5σ feature from a 1σ one. Second, an arcsinh RGB cannot represent negative numbers at all, which is *why* step (i) is needed in the first place — the display format forces the dishonesty.

(iv) Scale comparable to the seeing. The kernel $\sigma = 3\text{px}$ is $\approx 0.79''$, comparable to the $\sim 1.2''$ DECam PSF. A high-pass at the seeing scale subtracts real arc flux (the arc is itself only marginally resolved) and imprints a Mexican-hat ring around every compact source. Features in such a residual are partly manufactured by the filter, not present in the sky.

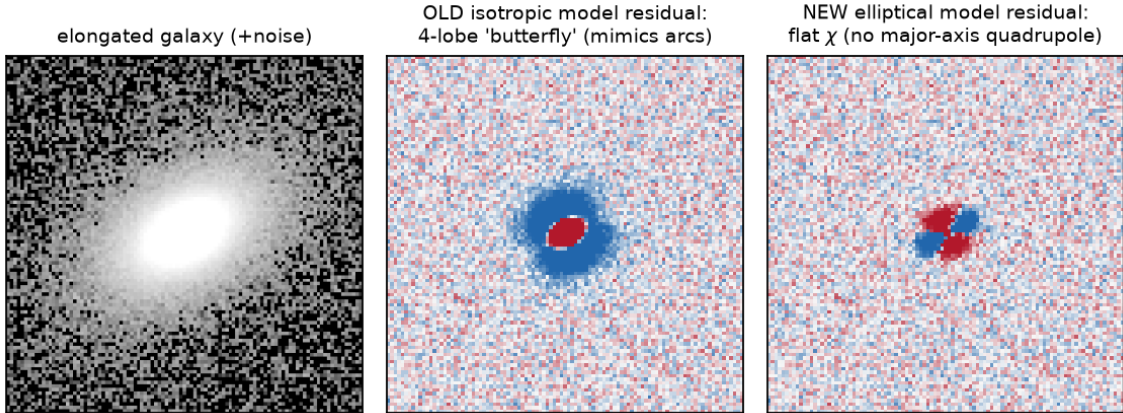


Figure 1: The butterfly artifact (failure ii), on a synthetic elliptical galaxy with noise. *Left:* the galaxy. *Middle:* the old isotropic-model residual data $-\mathcal{G}_{\sigma=3}[\text{data}]$, shown on the new signed scale at the same noise level — a strong central quadrupole (red along the major axis, blue along the minor), the lobes of which masquerade as arcs. *Right:* the new elliptical-model residual on the same data — flat noise, no major-axis quadrupole. Red is positive (excess), blue negative (over-subtraction).

3 The new transformation: a signed, noise-normalized elliptical residual

The replacement (`render.residual`) is built in three stages: model, noise, display (Fig. 2).

Model. For each band we subtract a smooth model of the galaxy formed as a *sigma-clipped median over elliptical annuli*. The galaxy’s center, ellipticity, and position angle are estimated once from the sigma-clipped second moments of the $r+z$ luminance; pixels are binned into one-pixel-wide elliptical annuli of that geometry, and each annulus is assigned its sigma-clipped median. Three properties make this honest. It carries the galaxy’s true ellipticity and orientation, so it leaves no butterfly. It is fit to, and subtracted in, the *seeing-convolved* data — there is no deconvolution, so the model is automatically PSF-matched, which matters because the cutouts carry no stored PSF. And because it is a *median*, a tangential arc that fills a minority of an annulus barely moves the model and therefore survives in the residual, whereas a least-squares or mean model would be dragged toward the arc and partly cancel it. The elliptical-annulus median is the azimuthally-resolved generalization of the circular profiles long used for galaxy photometry (Jedrzejewski, 1987), and is exactly the smooth-model recipe (median integration, 3σ clipping) the Siena Galaxy Atlas applies to Legacy Survey *grz* imaging at scale (Moustakas et al., 2023).

Noise. We divide by a robust per-band noise σ , the median-absolute-deviation standard deviation (`mad_std`) of the 3σ -clipped residual, cross-checked against the frame border; the sigma-clip rejects the arc and the galaxy core, so σ measures sky plus correlated noise. The displayed quantity is the dimensionless $\chi = (\text{data} - \text{model})/\sigma$.

Display. χ is rendered on a fixed $\pm 5\sigma$ red–blue diverging scale (red positive, blue negative, white at zero), arcsinh-compressed within the range, as a horizontal $g|r|z$ montage. Nothing is re-floored or clipped, so over-subtraction remains visible as blue. The fixed limits make every object directly comparable. The bands are shown separately because *cross-band coherence is the discriminant*: a

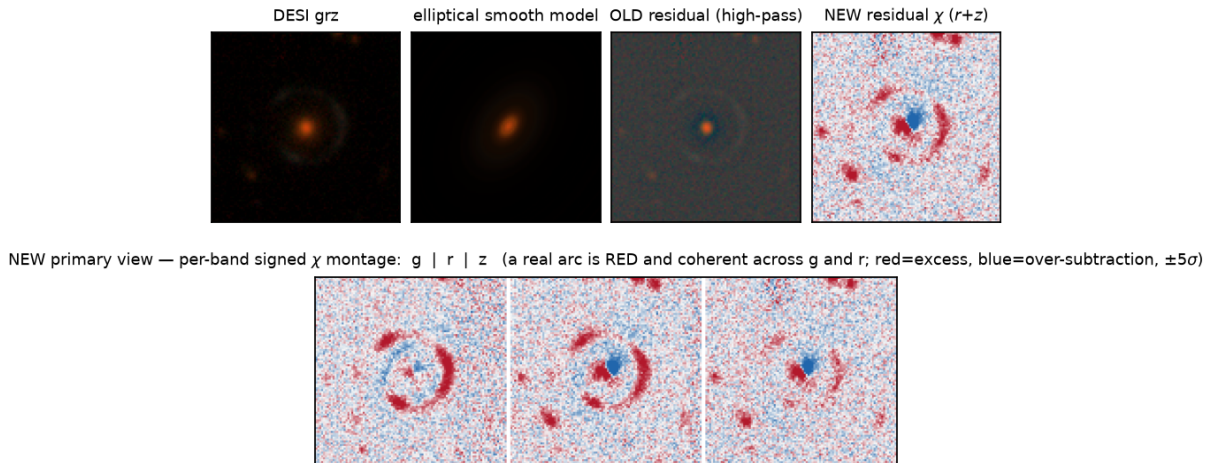


Figure 2: Anatomy of the new residual on the Cosmic Horseshoe (CSWA 1). *Top, left to right:* the DESI *grz* cutout; the elliptical smooth-light model that is subtracted; the old high-pass residual (washed out by the re-floor and arcsinh); and the new single-panel χ of the $r+z$ luminance. *Bottom:* the new primary view shown to the grader — the per-band signed- χ montage, in which the Einstein ring is red and coherent across g and r , with a blue (honestly displayed) over-subtracted core.

real lensed arc is a connected feature present in g and r (the source is blue, so it is strong in g/r and weak in z), whereas per-pixel noise and model-mismatch rings are not band-coherent. Crucially, the discriminant is *shape and cross-band coherence*, *not the residual’s sign*: because the source is blue and a single-elliptical subtraction can leave a real arc either red or blue, a viewer (or grader) must never equate “blue” with “artifact” — a lesson §7 learned the hard way. We never pass the residual through the Lupton transform, which cannot show negative values (Lupton et al., 2004).

4 Side-by-side comparison on real candidates

Figure 3 is the central comparison: the old and new residuals on the *same* DECam cutouts of four DR11 systems. The two rows of known lenses show the new residual revealing a red, cross-band-coherent arc (a near-complete ring for CSWA 1; a tangential arc for the Huang+2020 system), with the over-subtracted core honestly visible in blue. The two rows of **lrg+companion mimics** — candidates the finder ranked highly but which are not lenses — show the new residual surfacing the companion as a compact blob, *not* the tangential arc a genuine lens produces; the morphology, read across bands, is what separates them. In every row the old residual is a muddy blob with a high-pass ring around the core, conveying essentially nothing the grader can use. The same transformation that makes a real arc legible also refuses to manufacture one where the sky has only a companion galaxy.

5 Design choices, and what we deliberately did not do

Median model by default; isophote fitting as an opt-in. A floating elliptical-*isophote* fit (Jedrzejewski, 1987), which lets ellipticity and position angle vary with radius, is in principle more faithful and is available as an opt-in engine (`photutils.isophote` (Bradley et al., 2024)). In practice it converges on fewer than 15% of these faint 101-px *grz* cutouts and is 10–50 $\times$ slower, so on this data it would mostly fall back anyway; the elliptical-annulus median is the fast, always-converging

default, with the isophote engine reserved for higher signal-to-noise imaging (HSC/Euclid). A multi-Gaussian expansion (Cappellari, 2002) is a third, strictly-positive option in the same spirit.

Median over a more flexible fit — “under-fit on purpose”. A natural “improvement” is to fit low-order azimuthal harmonics per annulus, which removes the central PSF-circularization residual. It also *absorbs real arcs*: Figure 4 shows that on the Cosmic Horseshoe an aggressive harmonic fit eats the Einstein ring, while the median preserves it. Because every added degree of freedom can absorb a lensed feature (Peng et al., 2010), and because near-complete rings are exactly the most spectacular lenses, we deliberately under-fit: the median leaves an honest central residual rather than risk deleting the signal. This is the same lesson that motivates b-spline lens-galaxy subtraction in the SLACS program (Bolton et al., 2006) and the “no lens left behind” principle of *modelling* the source rather than masking it in forward fitting (Etherington et al., 2022). The rigorous endpoint — a full forward model of lens, source, and PSF with a χ residual (Birrer et al., 2021) — is the right tool for confirmation but is overkill for an unattended quick-look view over hundreds of cutouts.

6 Validation

Four synthetic tests (`tests/test_residual_honesty.py`), each building a PSF-convolved Sérsic galaxy ($\sim 1.2''$ FWHM) with noise matched to a real cutout, check the four properties the old residual violated; all pass. **(1) Noise consistency:** outside the inner $\sim 2''$ the residual χ is consistent with $\mathcal{N}(0, 1)$ (mean 0.00, standard deviation 0.99, 0.32% of pixels beyond 3σ versus the Gaussian expectation 0.27%), with no coherent structure — no butterfly. **(2) Arc non-absorption:** an injected tangential arc is recovered at 0.63 of its in-mask flux, against 0.08 for the old Gaussian high-pass. **(3) Sign preservation:** a deliberate 5% over-subtraction produces a clearly visible negative (blue) bowl that the old re-floor would have hidden (Fig. 5). **(4) Smooth-galaxy modelling:** on an elongated galaxy the elliptical model’s residual outside the core is $7\times$ smaller than the old high-pass’s, and is core-confined (it falls steeply with radius). The new residual costs about 40 ms per object and never fails to produce an image.

7 Does it change grading? A controlled A/B

The honesty tests above measure the residual against the *pixels*; they do not measure its effect on the downstream judgement. To do that we ran a controlled A/B that grades the *same* candidates twice, swapping only the residual view (old vs new) through a single environment lever (`LENSJUDGE_RESIDUAL_VERSION`, read by `render.render_views` and `residual_view_desc`); everything else — the full/zoom views, the rubric, the model — is held fixed. We graded 525 candidates with the deployed direct grader: 240 expert-graded A/B/C lenses, 130 Grade-D human-rejects (the hardest negatives), 130 random galaxies, and 25 spectroscopically-confirmed “gold” systems; two replicates per arm. The primary endpoint is the threshold-free ROC-AUC with a paired bootstrap CI; because the residual’s prompt wording shifts the grader’s operating point (see below), we also compare at a *matched* false-positive rate (Table 1).

It is discrimination-neutral. On the core detection task (lens vs random galaxy) the new residual does not measurably improve the grader: $\Delta\text{AUC} = +0.021$ with a 95% CI of $[-0.043, +0.085]$ that straddles zero (Fig. 6a), and at a matched FPR = 0.2 the grade-A recall is *identical* (0.256 in both arms). The large uncalibrated recall “gains” are a uniform threshold shift, not better separation

Table 1: The residual A/B (legacy old residual vs the new signed- χ residual; 525 candidates, two replicates per arm). AUC is the threshold-free primary endpoint with a paired bootstrap 95% CI; recall and FPR are at, or calibrated to, a matched operating point. The change is discrimination-neutral, preserves grade-A at a matched FPR, and needs calibration to control the false-positive flood.

quantity	legacy	new
AUC, lens vs random	0.650 [0.59, 0.70]	0.671 [0.61, 0.72]
Δ AUC (new–legacy)	+0.021 [−0.043, +0.085] (n.s.)	
AUC, lens vs Grade-D	0.387	0.463 (both < 0.5)
grade-A recall @ matched FPR= 0.2	0.256	0.256
random-galaxy FPR (raw $\rightarrow$ calibrated)	0.12 $\rightarrow$ 0.12	0.30 $\rightarrow$ 0.12
Grade-D FPR (raw $\rightarrow$ calibrated)	0.27 $\rightarrow$ 0.28	0.49 $\rightarrow$ 0.35

(below), and they vanish once the FPR is matched. On the hardest negatives (lens vs Grade-D) the new residual is significantly *less bad* (Δ AUC = +0.076, CI [+0.017, +0.140]), but both arms sit *below* 0.5: the grader still ranks human-rejected look-alikes above real lenses, so this is “less wrong,” not usable discrimination.

A regression we caught, and the lesson. The A/B exposed a real harm in the residual’s *prompt convention*, not its pixels. Our first deployed wording told the grader “red = arc, blue = over-subtraction artifact.” But a lensed source is blue, and after a single-elliptical subtraction a genuine arc often renders blue, so the grader — conflating the residual’s *sign* with the source’s *colour* — re-read real grade-A arcs as artifacts and rejected them (grade-A recall fell 0.259 $\rightarrow$ 0.185). Rewriting the description to be morphology-first (judge a *tangential*, *cross-band-coherent* feature offset from the over-subtracted inner $\sim 2''$ core; never treat blue as an artifact) restored grade-A non-inferiority. The residual’s prompt is load-bearing: it must key on shape and cross-band coherence, never on sign.

Hotter scores demand calibration. The new residual inflates p_{lens} roughly $2\times$ across the board; used raw it floods false positives (random-galaxy FPR 0.12 $\rightarrow$ 0.30, Grade-D 0.27 $\rightarrow$ 0.49; Fig. 6b) while the ROC-AUC barely moves — a pure operating-point shift. We therefore ship the new residual with an FPR-matched operating point (`eval/calibrate.py`: a p_{lens} threshold fit to a target FPR on a labelled negative set, persisted as `residual_op_point*.json`). Calibrated, the flood is removed and both arms sit at one operating point, where their grade-A recall coincides (Fig. 6c). The decision must use the calibrated threshold, never the raw grade.

What it is worth. The honest bottom line: the new residual is a *scientific-integrity* improvement that is *grading-discrimination-neutral*. Its benefits are a defensible, sign-preserving, noise-normalized image; rationales that key on real morphology; and decisive upgrades of clean tangential arcs — not a broad accuracy gain, and at $\sim 37\%$ more cost per call. Caveats: only two replicates per arm; grade-A ($n=39$) and gold ($n=21$) are underpowered, and gold’s matched-FPR recall trends down in a way the headline numbers hide; and the artifact-rejection failure was *reshaped*, *not removed* — faint, blobby, or close-pair real lenses are still sometimes dismissed as over-subtraction. These set the agenda for a larger, pre-registered follow-up.

8 Limitations

The honest residual is honest about its own limits. The cutouts carry *no PSF and no inverse-variance map*, so the model is only as PSF-matched as the blurred data (a faint central image inside the Einstein radius can still be partly absorbed), and the noise σ is *estimated*; because DESI coadds have spatially correlated (resampled) noise, χ slightly under-estimates the uncertainty of a multi-pixel arc and should be read as a display/triage significance, not a calibrated detection statistic. The single-ellipticity model leaves a central PSF-circularization residual confined to the inner $\sim 2''$, shown faithfully as a blue core that the grader is told to *ignore* rather than to read as an artifact verdict (the A/B of §7 showed an earlier “blue = artifact” instruction backfired); a galaxy that is genuinely irregular or strongly isophote-twisted will leave honest but structured residuals that a validity flag (e.g. **statmorph** (Rodriguez-Gomez et al., 2019)) can gate. Finally, a near-complete Einstein ring that fills most of an annulus is partly absorbed by *any* azimuthal-median model and is the case to escalate to a masked or forward-model fit. The imaging used here is the DESI Legacy Surveys DECam coadds (Dey et al., 2019); the same transformation is applied to the higher-resolution HSC and Euclid tier-2 residuals, where the isophote engine becomes viable. Finally, the grader’s raw p_{lens} on the new residual is uncalibrated and runs hot (§7), so any deployed decision must pass through the FPR-matched operating point, not the raw grade.

References

- S. Birrer, A. J. Shajib, D. Gilman, et al. lenstronomy II: A gravitational lensing software ecosystem. *Journal of Open Source Software*, 6(62):3283, 2021. doi: 10.21105/joss.03283.
- A. S. Bolton, S. Burles, L. V. E. Koopmans, T. Treu, and L. A. Moustakas. The Sloan Lens ACS Survey. I. A Large Spectroscopically Selected Sample of Massive Early-Type Lens Galaxies. *The Astrophysical Journal*, 638(2):703, 2006. doi: 10.1086/498884.
- L. Bradley, B. Sipőcz, T. Robitaille, et al. photutils: Photometry and related tools for Python, 2024. Astropy-affiliated package; `photutils.isophote` elliptical-isophote fitting.
- M. Cappellari. Efficient multi-Gaussian expansion of galaxies. *Monthly Notices of the Royal Astronomical Society*, 333(2):400, 2002. doi: 10.1046/j.1365-8711.2002.05412.x.
- A. Dey et al. Overview of the DESI Legacy Imaging Surveys, 2019.
- A. Etherington, J. W. Nightingale, R. Massey, et al. Automated galaxy-galaxy strong lens modelling: no lens left behind. *Monthly Notices of the Royal Astronomical Society*, 517(3):3275, 2022. doi: 10.1093/mnras/stac2639.
- R. I. Jedrzejewski. CCD surface photometry of elliptical galaxies. I. Isophotal analysis and structural parameters. *Monthly Notices of the Royal Astronomical Society*, 226:747, 1987. doi: 10.1093/mnras/226.4.747.
- R. Lupton, M. R. Blanton, G. Fekete, et al. Preparing Red-Green-Blue Images from CCD Data. *Publications of the Astronomical Society of the Pacific*, 116(816):133, 2004. doi: 10.1086/382245.
- J. Moustakas, D. Lang, A. Dey, et al. The Siena Galaxy Atlas 2020. *The Astrophysical Journal Supplement Series*, 269(1):3, 2023. doi: 10.3847/1538-4365/acfaa4.

- C. Y. Peng, L. C. Ho, C. D. Impey, and H.-W. Rix. Detailed Decomposition of Galaxy Images. II. Beyond Axisymmetric Models. *The Astronomical Journal*, 139(6):2097, 2010. doi: 10.1088/0004-6256/139/6/2097.
- V. Rodriguez-Gomez, G. F. Snyder, J. M. Lotz, et al. The optical morphologies of galaxies in the IllustrisTNG simulation: a comparison with Pan-STARRS observations. *Monthly Notices of the Royal Astronomical Society*, 483(3):4140, 2019. doi: 10.1093/mnras/sty3345.

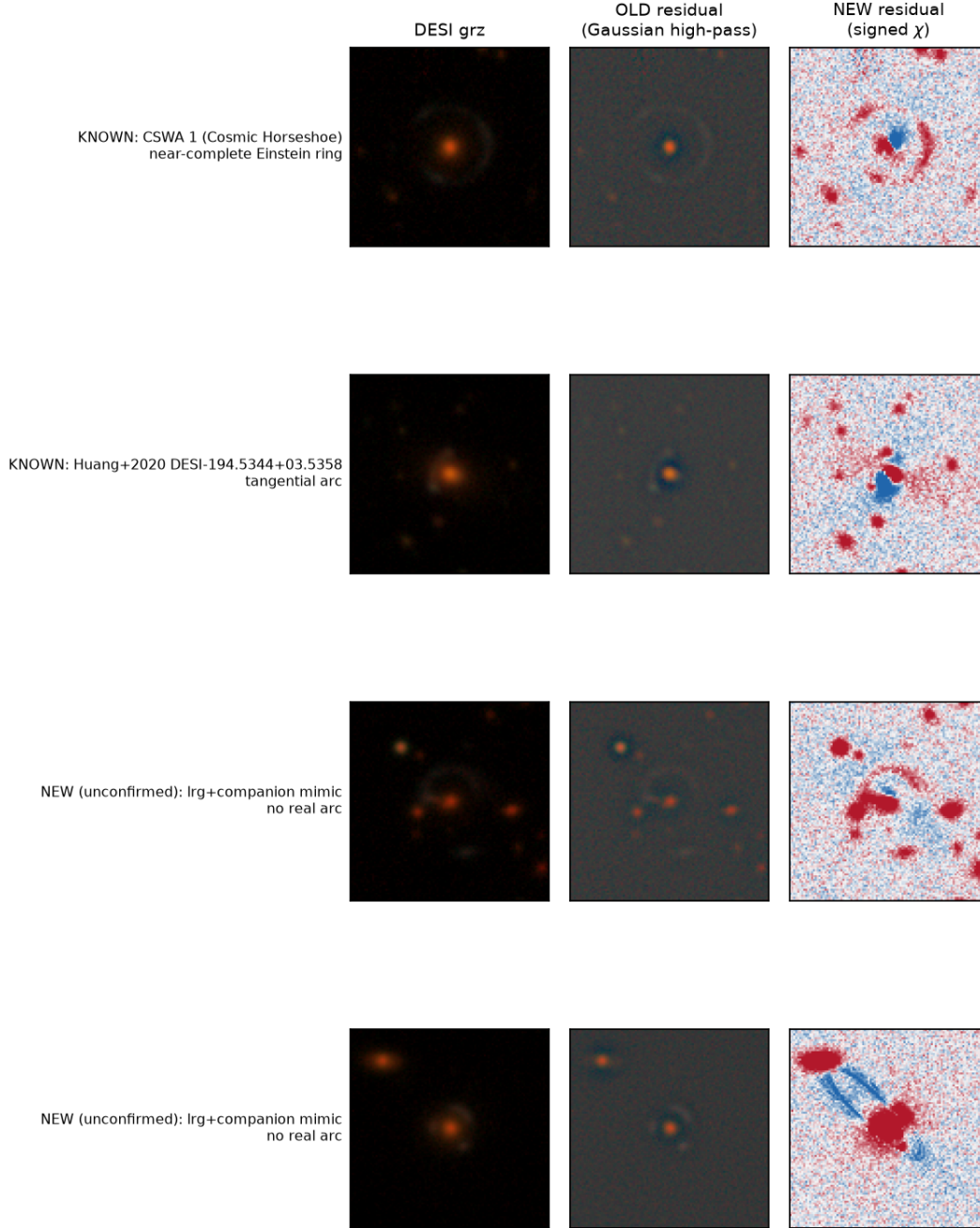


Figure 3: Old vs new residual on identical DESI cutouts (rows: two known lenses, two lrg+companion mimics). *Left:* DESI *grz*. *Middle:* the old Gaussian high-pass residual — uniformly muddy, with a ring artifact. *Right:* the new signed- χ residual ($r+z$ luminance) — a red arc for the lenses, only noise/blobs for the mimics. The new residual both *reveals* real arcs and *declines to fabricate* them.

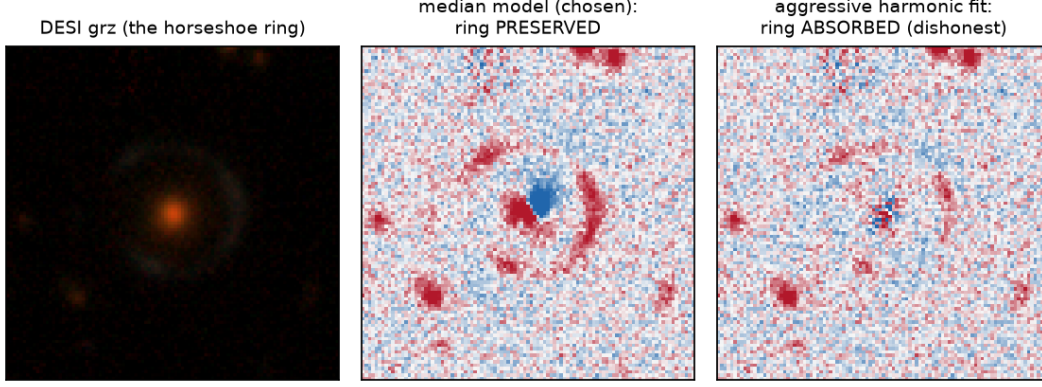


Figure 4: Why we under-fit. *Left:* the horseshoe in DESI *grz*. *Middle:* the chosen median model — the ring is preserved. *Right:* an aggressive per-annulus harmonic fit — the same ring is absorbed into the model and vanishes from the residual. A vanishing residual is a red flag, not a success.

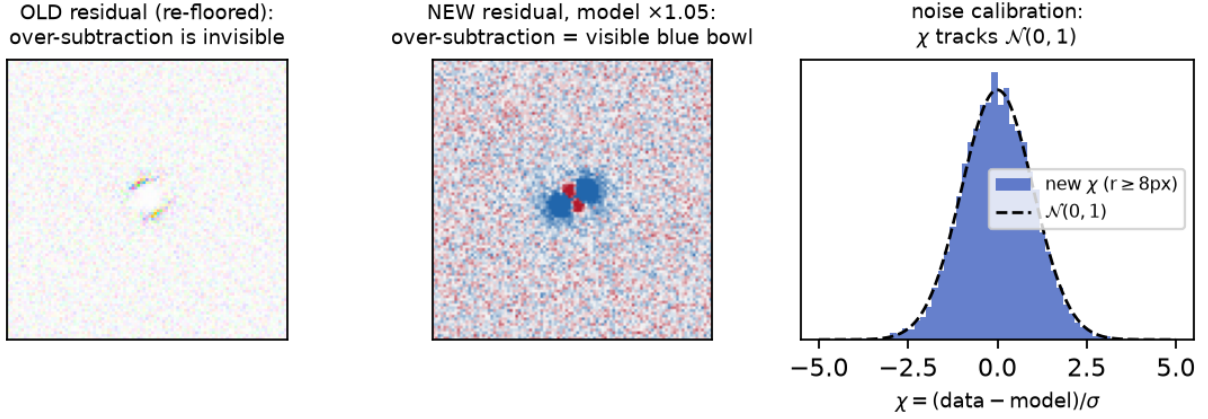


Figure 5: Sign and noise (validation tests 1 and 3). *Left:* the old residual re-floors, so an over-subtraction is invisible. *Middle:* the new residual of the same galaxy with the model scaled up 5% — the over-subtraction is a plainly visible blue bowl. *Right:* the new residual χ outside the inner core tracks $\mathcal{N}(0, 1)$, so a pixel value reads directly as a significance.

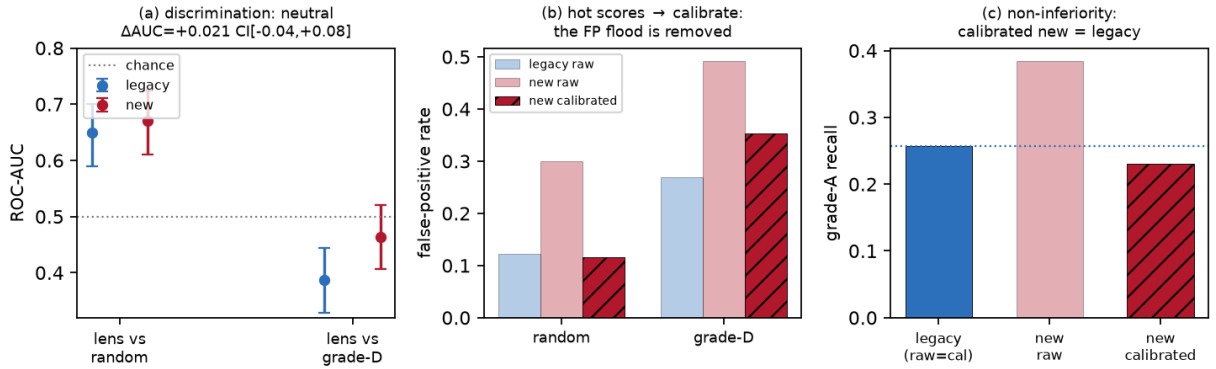


Figure 6: The grading A/B. (a) ROC-AUC with bootstrap 95% CIs: the arms overlap on lens-vs-random (the ΔAUC CI includes 0), and both are below chance on lens-vs-Grade-D. (b) Negative false-positive rate: the new residual's raw scores flood FPs; the FPR-matched calibration removes it. (c) grade-A recall: the new arm runs hot (raw), but calibrated it equals legacy — non-inferiority at the matched operating point.